

Fractal Information Theory Project

FIN ST342–ST356

Certified Fold Event, Symmetry Limits,
and Twelve-State Observer Depth

Local analytical, interval, exact finite, and bounded numerical research

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

August 17, 2026

Release 10.80

Repository snapshot. Audited tracked base: `bb540a767e210a5a3bf816e4868e7bcefbac3872`. The working tree contains uncommitted research artifacts; the accompanying SHA-256 ledger fixes this release packet.

License. CC BY 4.0. **Language.** English. **Resource type.** Preprint and executable reproducibility package.

Abstract

Programs ST342–ST356 continue the local FIN research program after Release 10.79. Two principal gaps are closed. First, the certified local coexistence bracket is narrowed from width 5×10^{-3} to 2×10^{-8} : outward endpoint objective intervals have opposite signs and one parametric Krawczyk tube contains a unique positive reflection-even stationary root across the full bracket. Second, the former numerical spinodal candidate is upgraded to a full 17-variable interval event certificate. The augmented stationarity–nullvector–normalization system has a unique root in the declared box, and the stationary Jacobian has exactly a one-dimensional kernel there. Conventional fold transversality coefficients remain point-certified only, so the claim is deliberately “certified fold event”, not a global bifurcation diagram.

A new obstruction explains why global closure remains difficult. D_{12} invariance maps minimizers into finite orbits but does not force them into reflection-fixed subspaces: the open dense simplex stratum is eleven-dimensional and has trivial stabilizer. A 173-start audit continues to find only the localized reflection orbit at $g = 4$, but this remains strong numerical evidence. Coupling spectrally truncated conservative mediators to the simplex shows that covariance permits ranks 1, 3, 5, 7, 9, 11; rank seven is the first at which a pure vertex beats the uniform state exactly at $g = 4$, and is also the first numerically localizing truncation.

The 12-state permutation-covariant confusion family admits an exact Le Cam depth: fine-to-coarse deficiency is zero and reverse deficiency is the difference of confusion errors. A strict-heat conditional Markov source yields an exact entropy rate and a numerical two-symbol rate–distortion curve. Two-cycle refinement complexes have two independent gauge-invariant holonomies, and multilevel clock errors separate into cancelling common calibration gauge and accumulating differential drift. No strict source for the active coupling or stabilizer coefficients, global physical vacuum, selector, absolute unit, time arrow, spacetime, apparatus, independent events, legacy-role transfer, Standard Model, gravity, L_{total} , or Theory-of-Everything closure is claimed.

Contents

1	Scope and proof discipline	1
2	Executive result ledger	1
3	ST342: a 2×10^{-8} coexistence bracket	2
4	ST343: full interval certification of the fold event	2
5	ST344–ST346: global symmetry limit and modal localization	3
6	ST347–ST350: invariants, pressure derivatives, and continuum design	4
7	ST351–ST353: scale cohomology and quantitative observer depth	5
8	ST354–ST356: strict-heat compression, clock drift, and evidence stop	5
9	Falsification ledger and present interpretation	6
10	Recommended programs ST357–ST371	7
11	Reproducibility and conclusion	8

1 Scope and proof discipline

The strict finite operator used in this batch is

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A\mathbf{1} = 0$ and $A \succeq 0$. This explicit diagonal convention is essential: the quoted row sum $s = 1.660307278766099$ belongs to the off-diagonal W .

The legacy ontological kernel remains an intermediate bridge object. No physical legacy role is silently transferred to the strict kernel. The internal ontology—the nadsoliton itself as primordial fractal information in a solitonic state, followed by light, matter, and an emergent observer—is preserved as project context but is not used to prove the finite theorems below.

[PROVEN] marks exact analysis or outward interval certification. **[STRONG NUMERICAL EVIDENCE]** marks reproducible floating computation. **[PARTIAL]** marks a missing global, continuum, or source step. **[BLOCKED]** marks evidence that cannot be generated locally.

2 Executive result ledger

Pro-gram	Status	Result
ST342	[PROVEN]	Local coexistence is certified in a width- 2×10^{-8} bracket.
ST343	[PROVEN]	Unique root of the full 17D augmented event system; stationary kernel dimension exactly one.
ST344	[PROVEN]	Symmetry alone cannot reduce all global minimizers to fixed-point subspaces.
ST345	[STRONG NUMERICAL EVIDENCE]	All 173 starts support the localized reflection orbit at $g = 4$; no global theorem.
ST346	[PROVEN]+[STRONG NUMERICAL EVIDENCE]	Rank seven is the first exact vertex threshold and first numerical localization among covariant modal mediators.
ST347	[PROVEN]	The spectral-energy subalgebra has 21 quartic and 56 sextic basis monomials; all 77 coefficients remain free.
ST348	[PROVEN]	Zero belief memory can coexist with nonzero Chernoff pressure derivative; contraction alone cannot close the derivative tail.
ST349	[PARTIAL]	Anisotropic widths improve the outer-cube pass count to 34/40, but six sections remain uncertified.
ST350	[PROVEN]+[STRONG NUMERICAL EVIDENCE]	Finite dual support and finite threshold-band structure are analytic; numerical grid endpoints remain uncertified.
ST351	[PROVEN]	A two-cycle refinement complex carries two independent gauge-invariant holonomies.
ST352	[PROVEN]	Exact Le Cam deficiency for the 12-state symmetric confusion family.
ST353	[PROVEN]	Matching $r\sigma^2/N$ recovery bounds in a declared spectral Gaussian experiment.
ST354	[PROVEN]+[STRONG NUMERICAL EVIDENCE]	Strict heat supplies a conditional Markov law and exact entropy rate; two-symbol rate-distortion is numerical.
ST355	[PROVEN]	Common scale gauge cancels while differential clock error accumulates by explicit products.
ST356	[BLOCKED]	No independently registered raw-event packet exists.

3 ST342: a 2×10^{-8} coexistence bracket

For reflection-even probabilities represented by seven orbit values q_i , the stationary equations at supplied active coupling g are

$$\log(12q_i) + 1 - g(Bq)_i + \lambda = 0, \quad q_0 + 2 \sum_{i=1}^5 q_i + q_6 = 1.$$

The new interval endpoints are

$$g_- = 2.902496471747767, \quad g_+ = 2.902496491747767.$$

At g_- , the complete objective enclosure is

$$[1.11419 \times 10^{-9}, 1.23114 \times 10^{-8}],$$

whereas at g_+ it is

$$[-1.23114 \times 10^{-8}, -1.11419 \times 10^{-9}].$$

Both endpoint roots are unique in their outward boxes. A coordinatewise parametric Krawczyk tube contains a unique positive root for every $g \in [g_-, g_+]$, with minimum inclusion margin 9.9883×10^{-8} .

Theorem 3.1 (narrow certified local coexistence). *The declared positive reflection-even branch has at least one point in $[g_-, g_+]$ at which its objective equals the uniform-state objective.*

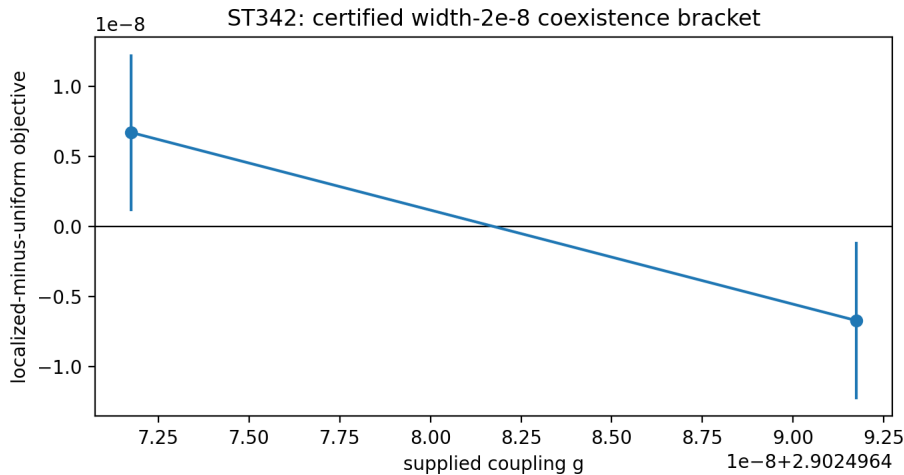


Figure 1: Outward endpoint objective intervals in the certified coexistence bracket.

This is a 250,000-fold width improvement over ST327. It still does not establish that the branch is globally minimizing, that the crossing is unique, or that g is internally sourced.

4 ST343: full interval certification of the fold event

Let $x = (q_0, \dots, q_6, \lambda)$ and define

$$\mathcal{F}(x, g, v) = \begin{pmatrix} F(x, g) \\ D_x F(x, g)v \\ (\langle v, v \rangle - 1)/2 \end{pmatrix} \in \mathbb{R}^{17}.$$

The previous floating candidate was at $g \approx 2.66490805100556$. ST343 evaluates all 17 residuals and the complete 17×17 Jacobian with outward interval arithmetic on a coordinate-radius 10^{-8} box. The Krawczyk image lies strictly inside, with minimum component margin 9.9988863×10^{-9} .

Theorem 4.1 (certified one-null-direction event). *The augmented system $\mathcal{F} = 0$ has exactly one root in the declared 17D box. At that root $D_x F$ has rank exactly seven and hence a one-dimensional kernel.*

The upper rank bound follows from the certified nonzero normalized nullvector. For the lower bound, the second-smallest singular value at the center is 3.38062, while the paid Frobenius perturbation bound over the box is below 7.1×10^{-5} , leaving rank-seven margin 3.38055.

The conventional point transversality diagnostics are

$$w^T F_g \approx -0.3572266, \quad w^T F_{xx}[v, v] \approx 0.2753332.$$

They strongly support a simple saddle-node fold, but were not enclosed as intervals in this batch. Accordingly the theorem certifies the event and nullity, not the entire local normal form, dynamic stability, or global branch diagram.

5 ST344–ST346: global symmetry limit and modal localization

The variational objective is invariant under the 24-element dihedral group D_{12} . This does not imply that every minimizer is reflection-even.

Theorem 5.1 (fixed-symmetry reduction no-go). *The union of nontrivial D_{12} fixed-point subspaces has empty interior in the 11-dimensional probability simplex. Its complement is open and dense and consists of trivial-stabilizer points with 24-element orbits. Hence symmetry alone cannot reduce the complete global minimization problem to reflection-fixed subspaces.*

An explicit witness is the normalized positive vector with entries proportional to $1, 2, 4, \dots, 2^{11}$: all entries are distinct, so no nonidentity dihedral permutation fixes it. Fixed-subspace searches remain legitimate for finding symmetric critical points; they are not an exhaustive global certificate.

ST345 therefore performs a falsification audit rather than claiming a proof. Across 173 uniform, vertex-biased, and random starts at $g = 4$, the best objective is -0.84519850768 , the largest probability is 0.99020346, and the minimum reflection defect is below 10^{-7} . Every best basin is numerically D_{12} -equivalent to the localized reflection orbit. This is compelling evidence, but ST344 explains why it is not enough.

For the mediator study, let A_m retain complete degenerate spectral blocks in descending eigenvalue order. Dihedral covariance permits only ranks

$$1, 3, 5, 7, 9, 11.$$

The pure-vertex energy crosses the uniform energy at

$$g_{E,m} = \frac{2 \log 12}{\text{tr}(A_m)/12}.$$

The values decrease from 25.4625 at rank one to 2.99331 at full rank. Rank seven is the first with $g_{E,m} < 4$, namely $g_{E,7} = 3.90777$, and is also the first truncation whose multistart optimum localizes numerically.

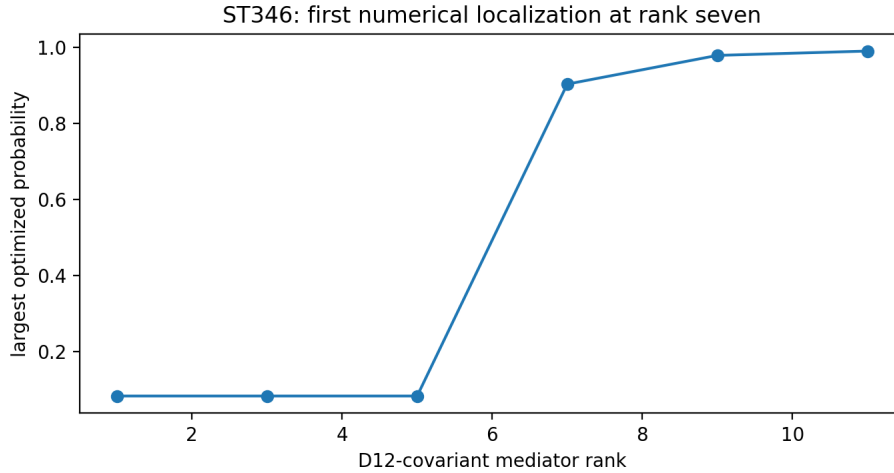


Figure 2: Uniform behavior through rank five and the first localized optimum at covariant rank seven.

Thus at least seven conservative hidden directions are needed before this particular simplex model can energetically prefer a vertex at $g = 4$. The result does not source rank, g , or a physical hidden system.

6 ST347–ST350: invariants, pressure derivatives, and continuum design

Write $E_k(q) = \|P_k q\|^2$ for the six positive real modal energies. Within the explicitly declared spectral-energy subalgebra, quartic invariants are $E_i E_j$ and sextic invariants are $E_i E_j E_k$.

Theorem 6.1 (spectral-energy invariant count). *The quartic basis has $\binom{7}{2} = 21$ monomials and the sextic basis has $\binom{8}{3} = 56$. The strict operator distinguishes weighted combinations such as $\langle q, Aq \rangle^2$ and $\langle q, Aq \rangle^3$, but does not choose the 77 general real coefficients or their signs.*

This scoped theorem is not a Molien classification of the full D_{12} invariant ring; phase-sensitive invariants can lie outside the modal-energy subalgebra.

ST348 proves that filter contraction cannot, by itself, close the Chernoff derivative problem. Consider one-state i.i.d. experiments

$$P = (0.9, 0.1), \quad Q = (0.6, 0.4).$$

Their belief-memory contraction is exactly zero, yet at $s = 1/2$ the normalized per-symbol pressure derivative is $0.0221383 \neq 0$. A valid all-depth theorem must therefore control the normalized multiplicative transfer weights, not only belief forgetting.

ST349 assigns coordinatewise half-widths from adjacent ST298 sections. The resulting anisotropy is real: 34 of 40 rigorous outer-cube tests pass, compared with 26 of 40 in the isotropic ST333 attempt. Six failures remain. This is useful method diagnosis: the next implementation must rotate coordinates into one longitudinal tangent and seven transverse directions rather than intervalize a large axis-aligned hull.

For ST350, finitely many response times t_j and one heat budget give a continuum linear program. Its dual uses finitely many nonnegative time weights α_j satisfying $\sum_j \alpha_j = 1$ and one heat price $\nu \geq 0$. The primal selector is determined by the sign of

$$\Theta(\mu) = \sum_j \alpha_j d_{t_j}(\mu) - \nu h(\mu).$$

On a compact positive spectral interval, a nonzero analytic Θ has finitely many zeros, hence finitely many threshold bands. The 250-, 500-, and 1000-point grids all find two selected bands (four entry/exit transitions) and value near 1.27633; exact continuum multipliers and endpoints remain uncertified.

7 ST351–ST353: scale cohomology and quantitative observer depth

For a connected refinement graph, logarithmic edge calibrations form one-cochains and vertex recalibrations are coboundaries. Therefore

$$\dim H^1 = E - V + 1.$$

The square-with-diagonal test complex has $V = 4$, $E = 5$, and two independent cycle holonomies. Direct vertex recalibration leaves both invariant to below 10^{-15} . This generalizes the one-cycle ST335 result without assigning physical scale curvature to FIN.

Let C_e be the 12-state permutation-covariant confusion channel with correct-label probability $1 - e$ and each incorrect label probability $e/11$.

Theorem 7.1 (exact twelve-state observer depth). *For $0 \leq e_f \leq e_c \leq 11/12$,*

$$\delta(C_{e_f}, C_{e_c}) = 0, \quad \delta(C_{e_c}, C_{e_f}) = e_c - e_f.$$

The finer experiment simulates the coarser by an additional symmetric confusion channel. Conversely, group-twirling cannot worsen worst-row total variation and reduces every candidate postprocessing to the one-parameter covariant family; identity is the boundary closest to the finer target. Full 144-variable stochastic LP replays agree with the formula to 5×10^{-16} for four test pairs.

For a known r -mode fiber in a Gaussian sequence experiment with orthonormal design, N_c independent counts and noise σ , least squares achieves

$$\mathbb{E} \|\hat{B} - B\|^2 = \frac{r\sigma^2}{N_c}.$$

The Gaussian translation minimax lower bound matches it. This is a sharp conditional result and clarifies the benefit of spectral restriction. It is not a detector model: known modal support, Gaussian noise, orthonormal design, and independence are premises.

8 ST354–ST356: strict-heat compression, clock drift, and evidence stop

At supplied $\tau = 0.5$, the strict heat kernel

$$P_\tau = e^{-\tau A}$$

is strictly positive and doubly stochastic; its smallest entry is 0.0107821. With uniform stationary distribution it defines a conditional, operator-generated 12-state Markov source. Its exact entropy rate is the common row entropy,

$$h(P_\tau) = 2.6057354210 \quad \text{bits per symbol.}$$

This is materially closer to FIN than the binary source used in ST338. A two-symbol Blahut–Arimoto computation gives the conditional Hamming rate–distortion curve below.

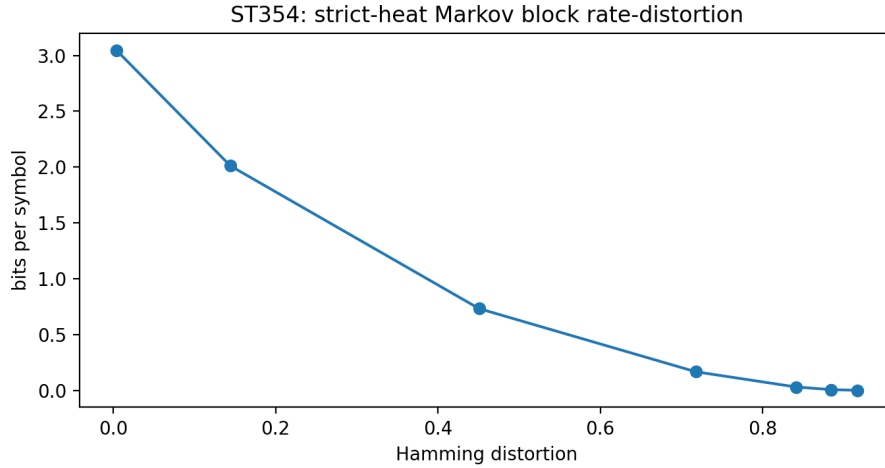


Figure 3: Two-symbol rate-distortion for the conditional strict-heat Markov source.

The operator supplies the transition only after the Markov reading and τ are declared. It does not identify physical entropy density, bits per volume, or compression of the universe.

Across several refinement levels, suppose mode i transforms as

$$\lambda_i^{(n+1)} = c_n \lambda_i^{(n)} (1 + \delta_{i,n}), \quad |\delta_{i,n}| \leq \eta_n < 1.$$

Every common calibration c_n cancels from a modal ratio. Differential unitary/heat clock drift obeys

$$\prod_n \frac{1 - \eta_n}{1 + \eta_n} \leq \frac{(\lambda_i/\lambda_j)_N}{(\lambda_i/\lambda_j)_0} \leq \prod_n \frac{1 + \eta_n}{1 - \eta_n},$$

and wave-clock ratios use square roots. This cleanly separates observer-layer gauge from real accumulated relative drift, but does not create seconds or an arrow of time.

ST356 applies the empirical stop rule. No independently registered, hashed raw-event file with separated provider, registrar, and analyst roles exists. Local code cannot generate that missing fifth measurement obligation.

9 Falsification ledger and present interpretation

Proposed conclusion	Falsification test	Surviving statement
The coexistence location is only a plot	Outward bisection and tube	False: a local crossing lies in a width- 2×10^{-8} interval.
The fold remains purely numerical	Full 17D Krawczyk test	False for event existence and nullity; transversality and global dynamics remain open.
Dihedral symmetry reduces the global problem	Trivial-stabilizer stratum	False: the generic simplex stratum remains 11-dimensional.
The reflection orbit is globally proved	173 starts plus ST344	Still false: strong evidence is not an exhaustive certificate.
Any low-rank mediator localizes	Covariant modal truncations	False at $g = 4$ through rank five; rank seven is first.
Strict invariance uniquely fixes a stabilizer	Quartic/sextic basis count	False: at least 77 coefficients are free in the scoped subalgebra.
Filter forgetting closes Chernoff asymptotics	i.i.d. zero-memory control	False: pressure derivatives can persist at zero memory.

Proposed conclusion	Falsification test	Surviving statement
Anisotropic widths close the branch tube	Outer interval tests	False so far: 34 pass and six fail.
Observer depth is only binary	12-state deficiency theorem	False in the symmetric family; the exact depth remains $e_c - e_f$.
Compression needs an unrelated source	Strict heat Markov law	Partly false: $e^{-\tau A}$ supplies a conditional source after τ and the Markov reading are declared.
Refinement calibration drift is inseparable	Product theorem	False: common gauge cancels; only differential error accumulates.
Local work is physical evidence	Independent record gate	False: ST356 remains blocked.

The deepest interpretation surviving this batch is:

Once a signed active interaction is supplied, FIN supports a sharply certified local variational geometry: localized coexistence is known to eight decimal places in the coupling, and a one-null-direction fold event is isolated in seventeen variables. The remaining obstacle is genuinely global and source-theoretic, not a lack of local numerical precision. Dihedral symmetry generates equivalent branches but cannot select or exhaust them. A sufficiently rich set of strict spectral modes can support localization, while operational observer depth, conditional compression, scale holonomy, and relative-clock drift admit exact quantitative laws. None of these laws supplies the active sign, its magnitude, an absolute unit, or an empirical realization.

10 Recommended programs ST357–ST371

- ST357.** Interval-enclose both fold transversality coefficients and derive the local saddle-node normal form from ST343.
- ST358.** Build entropy-convex cell lower bounds for a structure-aware global dual branch-and-bound, avoiding tensor grids.
- ST359.** Prove a D_{12} rearrangement theorem for the strict quadratic form or construct an explicit chiral lower-energy counterexample.
- ST360.** Conditional on ST358/ST359, certify or refute global $g = 4$ minimizer uniqueness modulo D_{12} .
- ST361.** Interval-certify the rank-seven localized state and compare it globally against uniform and boundary strata.
- ST362.** Compute the full Molien series and phase-sensitive degree-four/six D_{12} invariant basis beyond modal energies.
- ST363.** Prove a normalized Ruelle/Chernoff pressure-derivative bound for the frozen HMM or isolate a pressure-level counterexample.
- ST364.** Implement rotated tangent/transverse interval-Newton slabs and certify a continuous ST298 branch segment.
- ST365.** Interval-certify the continuum IR dual multipliers, all threshold zeros, and the corrected two-band optimum.
- ST366.** Extend ST352 from permutation-covariant confusion to arbitrary 12-state cyclic convolution instruments by a Fourier LP.

- ST367.** Replace the Gaussian fiber model by multinomial child counts and derive finite-sample confidence/minimax bounds.
- ST368.** Compute block lengths 3–5 for the strict-heat Markov source and bracket the asymptotic rate-distortion function.
- ST369.** Replace worst-case multilevel clock products by martingale/concentration bounds under a declared zero-mean refinement-error law.
- ST370.** Admit an active-coupling study only if a genuinely new strict typed source object is supplied; otherwise preserve the existing source no-go.
- ST371.** Resume empirical validation only after preregistered independent raw events and separated custody exist.

The highest priority is ST357 because it can convert the certified event into a complete local simple-fold theorem at low cost. ST358–ST360 address the harder global obstruction. ST362 and ST363 close clearly identified algebraic and information-theoretic gaps. ST370 is not permission to replay exhausted selector or bridge searches.

11 Reproducibility and conclusion

The executable source `fin_st342_st356_research.py` writes fifteen program packets, one aggregate JSON, a CSV ledger, and three figures. The independent suite `test_fin_st342_st356.py` performs sixteen live checks covering packet hashes, interval signs and inclusion, the 17D residual and rank margin, trivial stabilizer, nonpromotion of the global claim, modal thresholds, invariant counts, the Chernoff counterexample, incomplete slabs, IR dual normalization, cohomology rank, the full 12-state stochastic LP, sharp risk scaling, the strict-heat Markov law, multilevel drift, and the evidence stop. All sixteen tests pass.

Release 10.80 makes a real local advance: the phase-transition branch and its fold event are no longer merely high-quality plots. It simultaneously sharpens the principal negative conclusion. Better local certification cannot substitute for a global variational theorem or for the missing source of the active signed coupling. Those are now the two cleanly separated mathematical frontiers.